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# Challenges Identifying Newly Launched Objects

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**Abstract**

Whenever satellites are launched, one of the first challenges is to find and track all the objects associated with that launch. This process must be performed as expeditiously as possible, to support early-orbit operations and safety of flight (conjunction assessment). That tracking task normally falls to the Joint Space Operations Center (JSpOC), which must rely on radar or optical observations from the Space Surveillance Network and those observations do not include positive identification of the objects.

Large satellite operators often operate their own independent tracking networks, which can generate orbits with positive identification. Many small satellite operators—particularly cubesat operators—do not have an independent means of positively tracking their satellites. And even if they do, as the number of payloads per launch increases, the challenge of associating operator orbits with JSpOC orbits becomes more complicated—delaying the positive identification of all objects.

While those operators with independent tracking are able to perform early orbit operations, those that do not have independent tracking are at risk of losing their satellites, because they may not be able to locate them to perform vital early orbit tasks in time. In addition, since most conjunction assessment is performed using some JSpOC data, it may be impossible to screen for conjunctions (if the data has not been released), impossible to know which satellites are affected by a particular close approach (if the orbital data is available but the objects are unidentified), or simply create conflicting assessments of the true situation.

This paper discusses a simple technique which quickly assesses all available operator orbital data against the latest TLE data available from JSpOC. Results from this technique for the Indian PSLV-C37 launch with 104 payloads are compared to the tracking and identification performance from the PSLV-C34 launch with 20 payloads.

**Keywords:** (Space Situational Awareness)

## 1. Introduction

When Professors Jordi Puig-Suari of California Polytechnic State University and Bob Twiggs of Stanford University set out to design what are now known as cubesats, one of the challenges was to determine a minimum useful size for such a satellite. To be useful, one needed to be able to track these cubesats in orbit, otherwise it would be impossible to communicate with them. That meant cubesats had to be large enough to be tracked by the US Space Surveillance Network (SSN), which provides orbital data to the public.

Because the US government classifies the actual performance characteristics of military systems such as the SSN, Puig-Suari and Twiggs had to rely on a generic statement that the SSN could track objects the size of a softball or 10 cm in diameter. What this generic capability did not address were the abilities of individual radars, with different operating frequencies, to track objects over a variety of observing geometries and attitudes. Being able to 'see' a 10-cm object is not the same as being able to maintain track custody any time that object was above a specific radar's observing horizon. And it also didn't address the limitations of trying to use a radar to differentiate among dozens of objects of the same size and material characteristics orbiting in close proximity.

In recent years, we have seen several launches that deployed 30 or more cubesats at once. When that happens, the first challenge is to detect each of these objects and associate individual observations into tracks. The observations in a track are then used to generate an initial orbit. The next challenge is to detect these objects over other radars and associate the new observations with the right track. Misassociating observations with tracks results in bad orbits and makes both tracking and track association more difficult. Typically, the Joint Space Operations Center (JSpOC)—who collects observations and generates orbits from them—does not release orbital data (in the form of two-line elements or TLEs) until they feel the orbital quality is sufficient for reliable tracking, which can take days. Of course, with no orbital data, tracking by satellite operators can be difficult.

Once TLEs are released, the next challenge is to figure out which TLE goes with which object. Many cubesat operators use narrow-beamwidth, high-gain antennas to track the weak signals from their satellites. That means they can only track a small number of TLEs on any given pass. If they point their antenna at a TLE and hear nothing, does that mean they are tracking the wrong object, that the TLE is bad, or that their satellite is not transmitting? While the operator is trying to figure that out, they may be unable to perform key early-orbit operations vital to the survival of their satellite.

It can take days for the JSpOC to start releasing TLEs for a launch involving a large number of objects. And it can take even longer for them to identify each object, since they only have radar observations and do not have any way to monitor individual downlink frequencies that would help differentiate among these similar physical objects. Operators need to understand these limitations and be prepared to take actions that can accelerate the timeline for tracking and identifying their satellites. This paper will address some of those actions and show how they can combine with a simple tool to expedite identifying their payload(s).

## 2. Preparing for Success

There are a number of ways to lay the groundwork for improved tracking and identification of multiple objects deployed nearly simultaneously. Some are controlled by the launch provider while others are within the purview of the individual satellite operators.

The first thing to consider is how to provide JSpOC with a good starting point for track association. If JSpOC has good initial orbits to start with, associating initial observations with those 'tracks' can be much easier. Many launch providers make pre-launch orbital data available for each of their passengers. They may also provide post-deployment updates, as well. It is incumbent for operators to require their launch providers to make this information available—not just for their individual satellites but for everything being deployed. The data should be clearly defined—specifying coordinate frames, time systems, and units—and be in a fixed format that allows the data to be used in a consistent way for mission planning. There is little worse than taking the time to successfully develop, test, and validate a tool to convert the pre-launch data, only to find that the post-deployment data uses a different format, coordinate frame, or units—forcing the operator to try to adapt on the fly and possibly jeopardizing mission success.

And while there are any number of ways to describe an orbit, it is critical that those orbits be converted to a format that ensures interoperability with the JSpOC—which means converting them to TLEs that are used by their analysts and for tasking the SSN sensors. The conversion process requires fitting an orbit via a differential-correction process using a valid version of SGP4. Don't make the assumption that a TLE is merely Keplerian data laid out in a specific format—these are doubly-averaged mean elements and have to be produced in a way that is consistent with how they will be interpreted by SGP4. If you are unsure of how to do this, ask.

Proper pre-coordination among operators, the launch provider, and JSpOC can significantly improve the time to having good orbits released. Just be sure JSpOC gets the pre-launch data as soon as possible and understands what to expect regarding any post-deployment data.

Once this process is completed, focus shifts to identifying individual objects. Launch providers can help here, as well, by ensuring adequate separation at deployment. Simply ejecting everything from a stable inertial platform may result in a dense cloud of objects that takes time to spread out can make not only identification difficult, but complicate the track-association process. However, if the platform is still thrusting during deployment, this can provide cleaner separation of objects and facilitate both track association and identification. Be sure to ask your launch provider if this is an option.

But after all this is done, how exactly do you figure out which of the objects in this mass deployment belongs to you? Often operators search through the TLEs until they find one that 'works'—that is, allows them to communicate with their satellite(s). They may even report that information to JSpOC, which is great—except this approach frequently ends up with multiple operators claiming the same TLE. This approach can take an extended amount of time to finally identify all objects, as we will see later.

An efficient identification process requires the application of a matching or assignment algorithm that attempts to optimize some metric—perhaps the combined distances between each object and each TLE match. Of course, this requires some measurable data that can be compared to the orbital data. Ideally, it requires an independent means of tracking objects which have a known identification—something that is not currently available to many individual cubesat operators.

Some operators—notably Planet—have an independent way to range their satellites using the satellite telemetry. The advantage to this approach is that Planet knows the identification of each satellite they get ranging data for. Even though it will take some time to refine these orbits, track association will not be a problem. The same approach could be used by downlinking GPS data. And other, simpler, approaches could be applied—such as simultaneous Doppler tracking on multiple frequencies or even RFID tags flying through an activation beam to provide the independent observation side of this challenge. The cubesat community should seek ways to make it easy for all cubesats to have this capability.

The bottom line is that to enable expeditious tracking and identification requires informed operators that are aware of the limitations of current tracking systems and are prepared to work together to address them.

### 3. Methodology

So, let's look at some of the recent launches and ISS deployments involving Planet cubesats to first get a sense of the possible delays in tracking and identifying cubesats from large deployments. Then we'll describe a basic assignment application that can quickly associate operator orbital data with TLEs. We'll finish by seeing how well the application of that process to more recent launches has improved the overall result.

We'll start by looking at two sets of deployments—one from the ISS in May 2016 and another from the PSLV-C34 launch in June 2016—to see how long it took to reach each milestone in the identification process. We'll then look at what happened when we applied the process on the PSLV-C37 launch—with a record 104 satellites in February 2017—and then the KANOPUS-K-IV launch in July 2017. Table 1 shows the details of these 4 cases.

Table 1. Test Cases

| Pre-Development | | |
|---|---|---|
| Launch/Deployment | Total Payloads | Planet Payloads |
| ISS (2016 May 16-Jun 2) | 33 | 24 |
| PSLV-C34 (2016 Jun 22) | 20 | 12 |
| Post-Development | | |
| Launch/Deployment | Total Payloads | Planet Payloads |
| PSLV-C37 (2017 Feb 15) | 104 | 88 |
| KANOPUS-V-IK (2017 Jul 14) | 73 | 48 |

For each launch or deployment, we are looking for two key milestones for each object: when TLEs become available and when each object is first identified. TLEs may be generated, but there can be delays between generation and when the first TLEs show up on Space Track, so we track each of these milestones separately.

At some point, an entry will be added to the SATCAT (Satellite Catalog) associated with each object. Typically, when this first happens, each object will be identified as OBJECT X, where X is a sequential letter or letters indicating the order the object was added to the SATCAT. Since CelesTrak and JSpOC perform this task somewhat independently, times are tracked for each catalog. Finally, we also track once each object is uniquely identified in each catalog.

Tables 2 through 8 below summarize the results for each of the test cases. Each table shows the NORAD Catalog Number (often referred to as the SSC or Space Surveillance Center number) and current name for each object. Next to it is a timeline showing the occurrence of each milestone over a 5-week period starting just prior to the launch or deployment. The legend is found at the bottom of each table. Deployment or launch is shown in black, TLE availability in red, JSpOC SATCAT milestones in green, and CelesTrak SATCAT milestones in blue.

Let's begin by looking at the first test case, which demonstrates the challenges in tracking and identifying deployments from the International Space Station (ISS)—even though they may only involve a handful of cubesats at a time. This test case looks at 33 deployments from the ISS over the period from 2016 May 16 through June 2. Of those, 24 were Planet cubesats, which were independently tracked after deployment.

Upon reviewing the results, some of the problems become apparent very quickly. In almost all cases, there are delays of 1-2 days (or more) between deployment and TLE availability. For SSCs 41488-41490, there was a delay of 6 days. Obviously, it is impossible to track or identify an object without having a TLE first.

From that point, there are delays of days to weeks before JSpOC identified each object. In many cases, CelesTrak was able to make identifications much earlier by working directly with operators and assessing their means of identification. Planet is one of the operators that works with CelesTrak to identify their satellites, but Planet also lets JSpOC know of their identifications and provides both TLEs and ephemerides via their web site. Despite providing that information, there can still be significant delays in identifications in the JSpOC SATCAT.

Table 2. Test Case 1: ISS Deployments (2016 May 16-Jun 2)

| SSC/Name | Day 135-170 |
|---|---|
| Week | \| \| \| \| |
| 41474/MINXSS | A BCF D G E |
| 41475/CADRE | A BCF D G E |
| 41476/STMSAT-1 | A BDF GE |
| 41477/NODES 2 | A BDF GE |
| 41478/NODES 1 | A BDF GE |
| 41479/FLOCK 2E'-1 | A BDF G E |
| 41480/FLOCK 2E'-3 | A BDF G E |
| 41481/FLOCK 2E'-2 | A BDF G E |
| 41482/FLOCK 2E'-4 | A BDF G E |
| 41483/FLOCK 2E-1 | A BDF G E |
| 41484/FLOCK 2E-2 | A BDF G E |
| 41485/LEMUR-2-THERESACONDOR | A BCFD EG |
| 41486/FLOCK 2E-3 | A B CFD EG |
| 41487/FLOCK 2E-4 | A BC F DG E |
| 41488/LEMUR-2-NICK-ALLAIN | A B FCD EG |
| 41489/LEMUR-2-KANE | A BCF D EG |
| 41490/LEMUR-2-JEFF | A BCF D EG |
| 41563/FLOCK 2E-6 | A B D F G E |
| 41564/FLOCK 2E-5 | A BCFD G E |
| 41565/FLOCK 2E-7 | ABDF G E |
| 41566/FLOCK 2E-8 | ABDF EG |
| 41567/FLOCK 2E'-5 | A BCFD G E |
| 41568/FLOCK 2E'-6 | A B FD G E |
| 41569/FLOCK 2E'-8 | A B CFD G E |
| 41570/FLOCK 2E'-7 | A B CFD G E |
| 41571/FLOCK 2E-9 | A B CFD G E |
| 41572/FLOCK 2E-10 | A BCFD G E |
| 41573/FLOCK 2E-12 | A BC DF G E |
| 41574/FLOCK 2E-11 | A BC DF G E |
| 41575/FLOCK 2E'-9 | A BC DF G E |
| 41576/FLOCK 2E'-10 | A BC DF G E |
| 41577/FLOCK 2E'-11 | A BC DF G E |
| 41578/FLOCK 2E'-12 | A BC DF G E |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* |

The next case examines what used to be considered a large multi-payload launch—the PSLV-C34 launch of CARTOSAT-2C along with 19 other payloads. Of these, 12 were Planet cubesats. JSpOC did a good job of quickly getting TLEs out for the larger objects associated with the launch, but most of the Planet cubesats did not have any TLEs until 6 days following the launch. From there, it was several more days to get identifications—even for the larger payloads. For SSCs 41614 and 41615, it took 11 days from launch to JSpOC identification.

Table 3. Test Case 2: PSLV-C34 Launch (2016 Jun 22)

| SSC/Name | Day 170-205 |
|---|---|
| Week | \| \| \| \| |
| 41599/CARTOSAT-2C | A BC DF EG |
| 41600/SATHYABAMASAT | A BC DF E G |
| 41601/SKYSAT-C1 | A BC DF G E |
| 41602/GHGSAT-D | A BC DF EG |
| 41603/LAPAN-A3 | A BC DF E G |
| 41604/BIROS | A BC DF E G |
| 41605/M3MSAT | A BC DF EG |
| 41606/FLOCK 2P-6 | A BC DF EG |
| 41607/SWAYAM | A F BC DG |
| 41608/FLOCK 2P-11 | A F BC D EG |
| 41609/FLOCK 2P-2 | A F BC D EG |
| 41610/FLOCK 2P-9 | A F BC D EG |
| 41611/FLOCK 2P-4 | A F B C D EG |
| 41612/FLOCK 2P-10 | A F B C D EG |
| 41613/FLOCK 2P-8 | A F BC D EG |
| 41614/FLOCK 2P-12 | A F BC D G E |
| 41615/FLOCK 2P-7 | A F BC D G E |
| 41616/FLOCK 2P-5 | A F BC D EG |
| 41617/FLOCK 2P-1 | A F BC D EG |
| 41618/FLOCK 2P-3 | A F BC D EG |
| Codes | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* |

## 4. Tool Development

With upcoming multi-payload launches of as many as a hundred or more cubesats at a time on the horizon, it became evident that a more efficient means of making associations between operator data and JSpOC TLEs was needed. Even in the CelesTrak identifications, the process was manually intensive and could take hours to complete—making the process prone to error. What was needed was an automated tool that could take standard data sets, automatically assess the best assignment of JSpOC TLEs to operator data, and report that information to JSpOC in a standard way, to ease interpretation and build confidence in the results.

Manual analysis had shown that using a simple metric like range between the two data sets was insufficient to perform the assignment, since it failed to take into account relative motion. In fact, two orbits for an object might show physical proximity at some point in time but drift relative to each other, due to differences in semi-major axes. Experience had shown that orbits with close proximity and the same mean motion were more likely to be associated than those having the closest proximity at a specific analysis time.

As such, a tool was developed to calculate the root mean square (RMS) difference between each pair of orbital data sets from Planet and JSpOC over a number of orbits as the assignment metric. Since currency of the data was key to ensuring differences in drag modeling didn't throw off the results, the tool automatically retrieves the very latest data from both the Planet and Space Track web sites at the beginning of each analysis run.

Each analysis report summarizes the data in four ways. The first section shows the best mapping of each object in the operator data set to each object in the JSpOC data set for a particular launch. The next section shows the set of unique mappings of JSpOC data to the operator data. The next section shows those JSpOC TLEs with multiple allowable matches, ordered by increasing RMS. And the last section shows a list of JSpOC TLEs that had no matches in the operator data.

## 5. Tool Validation

Two sets of validation runs were performed to assess performance and selection criteria. The first assessed the full list of TLEs on the Planet web site (which at the time contained pre-launch TLEs for the upcoming PSLV-C37 launch of 88 cubesats) against the set of JSpOC TLEs for the entire operational Planet constellation. The second ran the same set of Planet TLEs against all ISS-related objects currently on orbit, by searching Space Track for TLEs associated with International Designator 1998-067.

In the first validation run, all matches were correct, with no Type I or Type II errors (false positives or false negatives). There were no cases of multiple possible assignments. And the 5 RAPIDEYE satellites (which are part of the Planet constellation but are not in the Planet TLEs) were positively identified as having no match. RMS values for matches ran from 0.1 km to 15.0

km. Most were within 2 km and the handful of larger RMS values were associated with ISS deployments nearing reentry. All expected non-matches had RMS values in the thousands of kilometers.

The second validation run produced similar results, with no Type I or Type II errors, no multiple possible assignments, and all non-Planet objects associated with the ISS correctly showing no match.

The tool was now ready for the PSLV-C37 launch and all we needed was data. JSpOC was apprised of our progress and asked to release TLEs as soon as possible, to aid in the identification process. Without any JSpOC data, we would be unable to perform any assignments or otherwise assist in the identification process. We expected that matches with poor RMS values would point to possible problems with the orbit solutions of either data set and that multiple assignments would suggest possible problems with track misassociation in the JSpOC data.

## 6. Operational Performance

The first set of JSpOC TLEs was released within a day of launch, but only for 7 objects (and one was for the rocket body). Since operator orbits were not yet available, matching was attempted using the ISRO-provided post-deployment state vectors converted to TLEs. Being able to get this data prior to launch and having time to develop tools to automatically convert it to TLEs was great, but format changes in the pre-launch data right before launch caused last-minute changes and additional testing to ensure we were ready.

Matching 104 objects against 6 TLEs—which were most likely for the largest objects from the launch—was not expected to produce useful results, particularly due to the close proximity of everything. Each TLE showed multiple possible assignments to 4-8 launch provider-based TLEs.

By the time that first analysis was completed, Planet reported that they had successfully contacted all 88 cubesats and performed two-way ranging on 30 of them, making it possible to switch to using their TLEs for further analysis.

We didn't receive any further JSpOC TLEs until 2017 February 17, when JSpOC sent us a set of 61 analyst TLEs to work with. The data format was slightly different than what we expected from Space Track, but we were able to make the necessary adjustments and perform the first good matching assessment. There were a handful of assignments that looked good, but many suggested more work on the orbit solutions was needed, which was expected.

While we did get updates for the first 6 public TLEs and were able to identify 2 matches and 2 TLE with multiple matches—each with a clear match for the first choice—we received no further analyst TLEs. One of the matches correctly (as we later realized) pointed out a misidentification of INS-1B, which was originally identified as SSC 41950. We received no additional public TLEs until February 21.

The milestone timelines in Tables 4 through 6 below show how things progressed from there. The CelesTrak identifications for SSCs 41954-41967 took about a week to assess, primarily because the RMS matches were not consistently below 5 km, where we expected they needed to be. With the release of TLEs for SSCs 41968-42017, our daily runs—which only took about a minute to perform—were showing consistent results day to day. Within a couple of days, we were able to confidently (and correctly) associate all the TLEs we had with Planet cubesats. By the time TLE data was released later that day for SSCs 42018-42042, we were able to make identifications that day or the next. When the final TLEs for SSCs 42043-42051 were released, we were able to identify those within a day, as well.

Even though we reported our results to JSpOC each day, it took a week or more after the CelesTrak identifications before they confirmed our results. In fact, a careful review of the tables below show that they still had not made some initial identifications until beyond 5 weeks after the launch. Admittedly, this was a new approach and our results probably didn't mesh well with their existing processes, so there was bound to be some challenges and likely skepticism. In the interim, we were able to work with other satellite operators like Spire and ISIS to identify their satellites, as well.

Looking back to past performance in identifying objects from other multiple-payload launches—like those in our first two case studies—suggests that we were able to significantly accelerate the identification process (whether JSpOC agreed initially or not). Had we had independent orbital data from all 104 payloads and JSpOC TLEs earlier, it seems likely performance would have been even better.

Table 4. Test Case 3: PSLV-C37 Launch (2017 Feb 15), Part 1

| SSC/Name | Day 45-80 |
|---|---|
| Week | \| \| \| \| |
| 41948/CARTOSAT-2D | AB FD E G |
| 41949/INS-1A | ABCFD E G |
| 41950/FLOCK 3P-20 | ABCFD E G |
| 41951/FLOCK 3P-8 | ABCFD G E |
| 41952/FLOCK 3P-51 | A BCFD G E |
| 41953/FLOCK 3P-37 | A BFD G E |
| 41954/INS-1B | A F BC D EG |
| 41955/FLOCK 3P-19 | A F BC D G E |
| 41956/FLOCK 3P-24 | A F BC D G E |
| 41957/FLOCK 3P-18 | A F BC D G E |
| 41958/FLOCK 3P-22 | A F BC D G E |
| 41959/FLOCK 3P-21 | A F BC D G E |
| 41960/FLOCK 3P-28 | A F BC D G E |
| 41961/FLOCK 3P-26 | A F BC D G E |
| 41962/FLOCK 3P-17 | A F BC D G E |
| 41963/FLOCK 3P-27 | A F BC D G E |
| 41964/FLOCK 3P-25 | A F BC D G E |
| 41965/FLOCK 3P-4 | A F BC D G E |
| 41966/FLOCK 3P-2 | A F BC D G E |
| 41967/FLOCK 3P-1 | A F BC D G E |
| 41968/FLOCK 3P-3 | A F BC D G E |
| 41969/FLOCK 3P-6 | A F BC D G E |
| 41970/FLOCK 3P-7 | A F BC D G E |
| 41971/FLOCK 3P-5 | A F BC D G E |
| 41972/FLOCK 3P-12 | A F B C D G E |
| 41973/FLOCK 3P-9 | A F BC D G E |
| 41974/FLOCK 3P-10 | A F BC D G E |
| 41975/FLOCK 3P-11 | A F BC D G E |
| 41976/FLOCK 3P-60 | A F BC D G E |
| 41977/FLOCK 3P-58 | A F BC D G E |
| 41978/FLOCK 3P-57 | A F BC D G E |
| 41979/FLOCK 3P-75 | A F BC D G E |
| 41980/FLOCK 3P-70 | A F BC D G E |
| 41981/FLOCK 3P-73 | A F BC D G E |
| 41982/FLOCK 3P-88 | A F BC D G E |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* |

Table 5. Test Case 3: PSLV-C37 Launch (2017 Feb 15), Part 2

| SSC/Name | Day 45-80 |
|---|---|
| Week | \| \| \| \| |
| 41983/FLOCK 3P-85 | A F BC D G E |
| 41984/FLOCK 3P-79 | A F BC D G E |
| 41985/FLOCK 3P-86 | A F BC D G E |
| 41986/FLOCK 3P-36 | A F BC D G E |
| 41987/FLOCK 3P-30 | A F BC D G E |
| 41988/FLOCK 3P-34 | A F BC D G E |
| 41989/FLOCK 3P-35 | A F BC D G E |
| 41990/FLOCK 3P-33 | A F BC D G E |
| 41991/LEMUR-2-SATCHMO | A F BC D G |
| 41992/LEMUR-2-MIA-GRACE | A F BC D G |
| 41993/LEMUR-2-SMITA-SHARAD | A F BC D G |
| 41994/LEMUR-2-SPIRE-MINIONS | A F BC D G |
| 41995/LEMUR-2-RDEATON | A F BC D G |
| 41996/LEMUR-2-NOGUECORREIG | A F BC D G |
| 41997/LEMUR-2-JOBANPUTRA | A F BC D |
| 41998/LEMUR-2-TACHIKOMA | A F BC D |
| 41999/BGUSAT | A F BC D |
| 42000/DIDO-2 | A F BC D G E |
| 42001/FLOCK 3P-49 | A F BC D G E |
| 42002/FLOCK 3P-67 | A F BC D G E |
| 42003/FLOCK 3P-68 | A F BC D G E |
| 42004/FLOCK 3P-41 | A F BC D G E |
| 42005/FLOCK 3P-45 | A F BC D G E |
| 42006/FLOCK 3P-48 | A F BC D G E |
| 42007/FLOCK 3P-43 | A F BC D G E |
| 42008/FLOCK 3P-42 | A F BC D G E |
| 42009/FLOCK 3P-61 | A F BC D G E |
| 42010/FLOCK 3P-40 | A F BC D G E |
| 42011/FLOCK 3P-16 | A F BC D G E |
| 42012/FLOCK 3P-14 | A F BC D G E |
| 42013/FLOCK 3P-53 | A F BC D G E |
| 42014/FLOCK 3P-54 | A F BC D G E |
| 42015/PEASSS | A F BC D G E |
| 42016/AL-FARABI 1 | A F BC D EG |
| 42017/NAYIF-1 (EO-88) | A F BC D G E |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* |

Table 6. Test Case 3: PSLV-C37 Launch (2017 Feb 15), Part 3

| SSC/Name | Day 45-80 | | | | |
|---|---|---|---|---|---|
| Week | | | | | |
| 42018/FLOCK 3P-23 | A F | | BCG D | E | |
| 42019/FLOCK 3P-76 | A F | | BCG D | E | |
| 42020/FLOCK 3P-69 | A F | | BCG D | E | |
| 42021/FLOCK 3P-84 | A F | | BC DG | E | |
| 42022/FLOCK 3P-59 | A F | | BC DG | E | |
| 42023/FLOCK 3P-32 | A F | | BC DG | E | |
| 42024/FLOCK 3P-71 | A F | | BCG D | E | |
| 42025/FLOCK 3P-77 | A F | | BCG D | E | |
| 42026/FLOCK 3P-80 | A F | | BC DG | E | |
| 42027/FLOCK 3P-66 | A F | | BC D | G E | |
| 42028/FLOCK 3P-65 | A F | | BC D | G E | |
| 42029/FLOCK 3P-50 | A F | | BCG D | E | |
| 42030/FLOCK 3P-52 | A F | | BCG D | E | |
| 42031/FLOCK 3P-46 | A F | | BCG D | E | |
| 42032/FLOCK 3P-47 | A F | | BCG D | E | |
| 42033/FLOCK 3P-44 | A F | | BCG D | E | |
| 42034/FLOCK 3P-64 | A F | | BCG D | E | |
| 42035/FLOCK 3P-63 | A F | | BCG D | E | |
| 42036/FLOCK 3P-62 | A F | | BCG D | E | |
| 42037/FLOCK 3P-38 | A F | | BCG D | E | |
| 42038/FLOCK 3P-39 | A F | | BCG D | E | |
| 42039/FLOCK 3P-15 | A F | | BCG D | E | |
| 42040/FLOCK 3P-13 | A F | | BCG D | E | |
| 42041/FLOCK 3P-55 | A F | | BC D G | | |
| 42042/FLOCK 3P-56 | A F | | BCG D | E | |
| 42043/FLOCK 3P-81 | A F | | | BC GDE | |
| 42044/FLOCK 3P-87 | A F | | | BC GDE | |
| 42045/FLOCK 3P-29 | A F | | | BC GD | |
| 42046/FLOCK 3P-82 | A F | | | BC GD | |
| 42047/FLOCK 3P-78 | A F | | | BC GDE | |
| 42048/FLOCK 3P-74 | A F | | | BC GDE | |
| 42049/FLOCK 3P-31 | A F | | | BC GDE | |
| 42050/FLOCK 3P-83 | A F | | | BC GD | |
| 42051/FLOCK 3P-72 | A F | | | BC GDE | |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* | | | | |

The next mass-deployment of Planet cubesats occurred just over 50 days prior to writing this paper, on 2017 July 14, on the KANOPUS-V-IK launch that included 48 Planet cubesats and 24 other payloads. In the intervening months, we had been able to tweak our tool's performance in anticipation of this launch. Unfortunately, we were not able to get pre-launch data to work with on this launch, so we would rely solely on Planet and JSpOC TLEs.

It took JSpOC several days to release any TLEs other than for KANOPUS-V-IK and the rocket body, but when they did, they released it all at almost the same time. That allowed us to identify all of the Planet cubesats with the exception of 2K-42 (a subject for another time) and 2K-06 (which didn't get its first TLE until a week later) within a couple of days, needed to assess matches and consistency of results over that time. And JSpOC agreed with/confirmed our results within a day. The results in Tables 7 and 8 for this launch show a dramatic improvement in time to identify payloads due to open data sharing, rapid analysis, consistent portrayal of results, and a transparent process that built confidence among Planet, JSpOC, and CelesTrak.

Table 7. Test Case 4: KANOPUS-V-IK Launch (2017 Jul 14), Part 1

| SSC/Name | Day 195-230 |
|---|---|
| Week | \| \| \| \| |
| 42825/KANOPUS-V-IK | A BCDEFG |
| 42826/NORSAT 1 | A BC FD E G |
| 42827/OBJECT C | A BC FD |
| 42828/NORSAT 2 | A BC FD E G |
| 42829/TECHNOSAT | A BC FD G E |
| 42830/OBJECT F | A BC FD |
| 42831/FLYING LAPTOP | A BC FD E G |
| 42832/OBJECT H | A BC FD |
| 42833/OBJECT J | A BC FD |
| 42834/OBJECT K | A BC FD |
| 42835/WNISAT 1R | A BC FD E G |
| 42836/OBJECT M | A BC FD |
| 42837/LEMUR-2-GREENBERG | A BC FD G E |
| 42838/LEMUR-2-ANDIS | A BC FD G E |
| 42839/LEMUR-2-MONSON | A BC FD G E |
| 42840/LEMUR-2-FURIAUS | A BC FD G E |
| 42841/LEMUR-2-PETERG | A BC FD G E |
| 42842/LEMUR-2-DEMBITZ | A BC FD G E |
| 42843/OBJECT U | A BC FD |
| 42844/NANOACE | A BC FD EG |
| 42845/LEMUR-2-ZACHARY | A BC FD G E |
| 42846/CORVUS BC2 | A BC FD E G |
| 42847/CORVUS BC1 | A BC FD E G |
| 42848/OBJECT Z | A BC FD |
| 42849/OBJECT AA | A BC FD |
| 42850/FLOCK 2K-03 | A BC FD G E |
| 42851/FLOCK 2K-04 | A BC FD G E |
| 42852/FLOCK 2K-01 | A BC FD G E |
| 42853/FLOCK 2K-02 | A BC FD G E |
| 42854/FLOCK 2K-47 | A BC FD G E |
| 42855/FLOCK 2K-48 | A BC FD G E |
| 42856/FLOCK 2K-45 | A BC FD G E |
| 42857/FLOCK 2K-24 | A BC FD G E |
| 42858/FLOCK 2K-46 | A BC FD G E |
| 42859/FLOCK 2K-23 | A BC FD G E |
| 42860/FLOCK 2K-21 | A BC FD G E |
| 42861/FLOCK 2K-22 | A BC FD G E |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID,*<br>*F = 1st SATCAT, G = 1st SATCAT ID* |

Table 8. Test Case 4: KANOPUS-V-IK Launch (2017 Jul 14), Part 2

| SSC/Name | Day 195-230 |
| --- | --- |
| Week | \| \| \| \| |
| 42862/FLOCK 2K-07 | A BC FD G E |
| 42863/FLOCK 2K-08 | A BC FD G E |
| 42864/FLOCK 2K-05 | A BC FD G E |
| 42865/FLOCK 2K-40 | A BC FD G E |
| 42866/FLOCK 2K-39 | A B C FD G E |
| 42867/FLOCK 2K-37 | A B C FD G E |
| 42868/FLOCK 2K-38 | A BC FD G E |
| 42869/FLOCK 2K-31 | A BC FD G E |
| 42870/FLOCK 2K-32 | A BC FD G E |
| 42871/FLOCK 2K-29 | A B C FD G E |
| 42872/FLOCK 2K-30 | A BC FD G E |
| 42873/FLOCK 2K-44 | A B C FD G E |
| 42874/FLOCK 2K-43 | A B C FD EG |
| 42875/FLOCK 2K-41 | A BC FD G E |
| 42876/FLOCK 2K-36 | A B C FD G E |
| 42877/FLOCK 2K-35 | A B C FD G E |
| 42878/FLOCK 2K-34 | A B C FD G E |
| 42879/FLOCK 2K-33 | A BC FD G E |
| 42880/FLOCK 2K-28 | A B C FD G E |
| 42881/LEMUR-2-ARTFISCHER | A B C FD G E |
| 42882/FLOCK 2K-27 | A BC FD G E |
| 42883/FLOCK 2K-26 | A BC FD G E |
| 42884/FLOCK 2K-25 | A BC FD G E |
| 42885/FLOCK 2K-20 | A BC FD G E |
| 42886/FLOCK 2K-19 | A BC FD G E |
| 42887/FLOCK 2K-18 | A BC FD G E |
| 42888/FLOCK 2K-17 | A BC FD G E |
| 42889/FLOCK 2K-16 | A BC FD G E |
| 42890/FLOCK 2K-15 | A BC FD G E |
| 42891/FLOCK 2K-13 | A B C FD G E |
| 42892/FLOCK 2K-14 | A BC FD G E |
| 42893/FLOCK 2K-12 | A BC FD G E |
| 42894/FLOCK 2K-11 | A BC FD G E |
| 42895/FLOCK 2K-10 | A BC FD G E |
| 42896/FLOCK 2K-09 | A BC FD G E |
| 42897/FLOCK 2K-06 | A B CDEFG |
| *Codes* | *A = Deployment, B = 1st TLE, C = 1st TLE Release, D = 1st SSR, E = 1st ID, F = 1st SATCAT, G = 1st SATCAT ID* |

## 7. Summary

These results show that it is possible to significantly improve the timeline for the release of orbital data and identification of that data in support of early orbit operations and safety of flight for large multi-payload deployments. The improvement is possible as a result of open collaboration and transparency, along with standard applications, data formats, and reporting of results. These types of improvements are necessary to support the growing demand for these launches.

It is also clear that when operator data was not available for other payloads, associations of TLEs with specific objects was significantly delayed. Quick identification of the JSpOC TLEs for the Planet cubesats did narrow the list of options for other operators, but that did not seem to help much. In at least one case, we had to tell an operator that their identification was incorrect, because we had already identified that TLE with a Planet cubesat with high confidence. That also proved to be the case with the early misidentification of INS-1B.

The cubesat community needs to work together to develop and encourage use of independent tracking mechanisms for all cubesats to facilitate this process, in the interest of mission assurance and safety of flight. And the community as a whole—launch providers, satellite operators, and space surveillance centers—needs to be open to sharing and collaboration in support of these goals.